

FINAL CAPSTONE REPORT

Edmonton Residential Real Estate: Data Engineering, Machine Learning &
Forecasting

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1. Introduction

1.1. Background of the Problem

Predicting real estate prices has become increasingly important for investors, policymakers, and homeowners. Accurate price forecasts help governments understand market stability, assist developers in planning housing supply, and guide individuals making long-term investment decisions. However, despite advances in machine learning and statistical modelling, many real estate forecasting models fail to produce reliable predictions. The failure of these models is rarely due to weak algorithms. Instead, the primary issue lies in the quality and structure of the data used for modelling. Real estate datasets often suffer from fragmentation, inconsistent identifiers, and administrative changes that break the continuity of historical records. When these issues remain unresolved, predictive models are trained on incomplete or misleading data, producing forecasts that do not accurately reflect market dynamics.

This project addresses that foundational problem directly. By designing and implementing an eight-stage transformation pipeline for the Edmonton residential real estate dataset (2000–2025), the team converted 515,122 raw transaction records into a 492,793-record modeling-ready dataset. Three machine learning models were subsequently trained and evaluated, and a SARIMA time-series comparison was conducted to validate the impact of data reconciliation on forecast accuracy. The final deliverables include a fully documented notebook pipeline, an output prediction dataset, an interactive Streamlit dashboard, and a Power BI report — all consistent with the artifacts described in this document.

1.2. The Problem of Broken Data

The Edmonton real estate dataset contains structural inconsistencies that disrupt the continuity of property histories. The most significant is LINC identifier drift. The Land Identification Number Code (LINC) is intended to serve as a stable primary key for each property. However, administrative changes — land subdivision, parcel consolidation, condominium conversion — cause the identifier to change. As a result, the same physical property appears as multiple unrelated assets, and long-term price histories disappear from the dataset entirely.

Additional structural issues identified include:

- Changes in LINC identification numbers due to land subdivision or consolidation
- Neighbourhood renaming — Oliver officially renamed to Wìhkwêntôwin, affecting 9,357 records
- Pre-construction ghost asset records (9,246 records) with FlrArea = 0 recorded alongside completed homes
- Address formatting inconsistencies preventing proper record matching across time periods

1.3. Research Questions

RQ1: Does reconnecting fragmented property histories improve the measurement of long-term appreciation?

RQ2: Are Edmonton property prices primarily driven by macroeconomic forces or local structural changes?

RQ3: How much does forecast accuracy improve when SARIMA models are trained on reconciled versus raw fragmented data?

1.4. Project Scope

- Cleaning and reconciliation of 515,122 real estate records spanning 25 years (2000–2025)
- Identification and isolation of 9,246 speculative pre-construction ghost asset transactions
- Standardization of address fields and neighbourhood identifiers
- Integration of 1,362 weekly Bank of Canada mortgage rate observations
- Engineering of five advanced analytical features (F1–F5) aligned to all three research questions
- Training and comparative evaluation of three ML models: OLS, Random Forest, Gradient Boosting
- SARIMA(2,1,2)(1,1,1,12) forecasting comparison on reconciled versus fragmented data
- 5-layer offline geocoding pipeline achieving 100% spatial coverage across all records
- Interactive Streamlit dashboard and Power BI decision-support report

2. Research Questions

2.1. Research Question 1: Impact of Historical Reconnection

Does reconnecting fragmented property histories improve the measurement of long-term appreciation?

A major limitation of real estate datasets is the fragmentation of property histories. In Edmonton, property identifiers frequently change due to administrative processes. When these changes occur, earlier sales records become disconnected from the property's current identity, creating artificial discontinuities. Properties that have existed for decades appear to have very short historical timelines, which significantly distorts long-term price analysis.

To address this, a custom reconciliation pipeline was developed using a standardized Base_Address field that removes unit numbers and suite identifiers from raw address strings. By focusing on the physical street address, the pipeline links multiple LINC numbers associated with the same property. The reconciliation resolved 234,486 drift instances, restoring 370,159 previously severed repeat transaction connections representing 75.1% of the final dataset.

Transaction Metric	Value
Total sales records processed	492,793
Repeat transactions restored	370,159
Historical reconnection rate	75.11%
Unique valid property identifiers	122,634
LINC drift instances resolved	234,486 (65.66% efficiency)
Near-duplicate records removed	22,311 (< 90 days, < 2% price change)

2.2. Research Question 2: Macroeconomic vs. Local Drivers

Are Edmonton property prices primarily driven by macroeconomic forces or local structural changes?

To differentiate between these drivers, a Price Ratio metric was introduced: Sold Price divided by List Price. Transactions with ratio ≤ 1.25 are classified as Macro-Driven; those above 1.25 as Internal (Property-Specific). Historical mortgage rate data from the Bank of Canada (Series V80691335) was merged using pandas merge_asof, ensuring each sale is matched to the most recent available rate with no temporal leakage.

The Macro segment (490,487 transactions, 99.5% of the dataset) shows a Pearson R of -0.24 with mortgage rates, confirming that interest rates act as a price ceiling for the majority of Edmonton real estate transactions. OLS regression confirmed the Mortgage_Rate coefficient at -0.116 ($p < 0.001$), and ANOVA across eight market cycle phases produced $F = 16,194.68$ ($p < 0.001$, Cohen's $f = 0.484$ — large effect). Tukey HSD post-hoc tests confirmed all 28 phase pairs are significantly different.

2.3. Research Question 3: Forecast Accuracy Validation

How much does forecast accuracy improve when SARIMA models are trained on reconciled versus raw fragmented data?

Two SARIMA models were constructed under identical conditions. An AIC grid search over all SARIMA(p,1,q)(1,1,1,12) combinations for p,q {0,1,2} identified (2,1,2) as the optimal order (AIC = 5,452.71). Both models were fitted using this order and evaluated on the 2024–2025 holdout period.

Model	Training Data	AIC	Performance
Random Walk Baseline	—	\$50,076	Baseline
Seasonal Naïve	—	\$23,652	Baseline
SARIMA Model A	Raw / Fragmented	\$37,500	+25.1% better than RW
SARIMA Model B	Reconciled + Inflation-Adj.	\$47,260	+5.6% better than RW

Model B underperforms Seasonal Naïve in the stable 2024–2025 test period. This is an honest and acknowledged finding: when prices are near-flat and seasonal patterns dominate, year-over-year prediction is competitive. Both models outperform the random walk benchmark. The 2026 annual forecast from Model B is \$415,555–\$448,260 (constant 2024 dollars, 95% CI: \$395,075–\$496,493).

3. Methodology

3.1. Analytical Pipeline

The analytical pipeline converts raw compressed datasets into a unified master dataframe suitable for modelling. The pipeline consists of eight sequential stages covering extraction, ingestion, cleaning, feature engineering, macroeconomic integration, geocoding, and time-series modelling.

Stage #	Process	Issues Addressed	Results
1	Data Ingestion	Multi-system encoding corruption	515,122 records loaded; latin1 encoding
2	Column Standardization	Special-character KeyErrors	All 57 columns renamed and validated
3	Datetime Coercion	Corrupt dates (e.g. year 2992)	18 corrupt rows removed; 515,104 valid
4	Neighbourhood Crosswalk	Oliver → Wîhkwêntôwin split as two regions	9,357 records reclassified; 1,068 neighbourhoods
5	Base_Address Engineering	234,486 LINC drift instances	122,634 unique physical parcels identified
6	Ghost Asset Classification	9,246 pre-construction records FlrArea = 0	Reclassified as Pre-Construction/Land
7	5-Layer Geocoding	No spatial coordinates in raw sold CSV	100% coverage; Geo_Source + Geo_Confidence added
8	BoC Mortgage Rate Sync	No macroeconomic context in raw data	1,362 weekly observations merged; 0 nulls

3.2. Methodological Validation: Data Quality Stress Test

To justify the extensive reconciliation and Base_Address engineering phases, a comparative stress test was conducted. Two independent SARIMA models were trained under identical hyperparameter conditions to isolate the impact of data integrity on predictive accuracy. Model A (Broken) was trained on the raw fragmented dataset; Model B (Reconciled) on the cleaned master dataset. The results confirm that without reconciliation, the model inherits structural noise that produces artificially smooth but misleading confidence intervals.

3.3. Feature Engineering

Five advanced engineered features were created beyond the cleaning pipeline, each designed to directly operationalize one or more research questions.

Feature ID	Variable(s)	RQ	Transformation
F1	Price_Ratio, Primary_Driver	RQ2	Ratio = Sold/List. Threshold 1.25 → Macro vs Internal
F2	Rolling_3Yr_Neighbourhood_Growth	RQ1/2	36-month rolling median % change per neighbourhood. 296,032 records, 86% coverage
F3	Market_Cycle_Phase, Macro_Context	RQ2/3	8 named BoC rate regimes verified against actual weekly BoC CSV data
F4	Is_Repeat, Days_Since_Last_Sale, Appreciation_%	RQ1	groupby().shift() on Base_Address. 370,159 restored 22,311 dupes removed
F5	Inflation_Adjusted_Price, Real_Price_Index	RQ3	CPI deflation: Sold × (CPI_2024/CPI_year). Base year 2024 = 124.5. Stats Can 18-10-0004-01

F1 — Price_Ratio and Primary_Driver: Ratio = Sold_Price / List_Price. Threshold 1.25 separates macro-driven transactions (≤ 1.25 , 490,487 records) from internal renovation-driven ones (> 1.25 , 2,414 records). Pearson R = -0.24 in the macro segment confirms interest rates act as a price ceiling.

F2 — Rolling_3Yr_Neighbourhood_Growth: Monthly median sold price computed per neighbourhood, then a 36-month rolling median applied. The 3-year growth rate is merged back on Community + Month. Range: -53.8% to +233.0%, Median: +7.0%. Fragmented data produces incomplete rolling windows; reconciled data fills them.

F3 — Market_Cycle_Phase and Macro_Context: Eight named BoC rate regimes from Post-Dot-Com High Rate (2000–2002) through Normalizing Market (2024–2025). A simplified Macro_Context (High Rate / Low Rate / Current Market) is derived for direct model input. Rate statistics verified against actual loaded BoC CSV data.

F4 — Repeat-Sale Features: groupby().shift(1) on Base_Address retrieves the prior sale. Days_Since_Last_Sale and Appreciation_Since_Last_Sale_Pct computed for all repeat transactions. Records within 90 days at < 2% price change removed as data-entry duplicates. Appreciation capped at $\pm 500\%$.

F5 — Inflation_Adjusted_Price and Real_Price_Index: Nominal prices over 25 years conflate genuine appreciation with inflation. From 2000 to 2025, nominal median prices rose +189% but real prices only +73% — confirming ~60% of apparent appreciation was inflation rather than structural market growth.

3.4. Geocoding Strategy

The midterm submission demonstrated geocoding using the Nominatim OpenStreetMap API, which produced a 42% failure rate on a 1,000-record sample and would require approximately 143 hours of runtime for the full dataset. For the final submission, a fully offline 5-layer fallback geocoding pipeline was integrated directly into Stage 7, achieving 100% spatial coverage with no API calls.

Layer	Method	Records	Confidence
L1	Exact LINC → assessment file join	6,568	HIGH — point-level GPS
L3A	Postal code → assessment file median coordinates	Varies	MEDIUM — postal code level
L3B	Postal FSA → Statistics Canada centroid dictionary (51 Edmonton FSAs)	~404,452	MEDIUM — FSA zone level
L4A	Neighbourhood → assessment file median coordinates	20,926	LOW — neighbourhood level
L4B	Neighbourhood → City of Edmonton boundary polygon centroid	10	LOW — neighbourhood level
L5	Edmonton city centre default (53.5461, -113.4938) — last resort	83,148	IMPUTED — city level only

Two metadata columns — Geo_Source and Geo_Confidence — document exactly how each record was geocoded, enabling downstream models to filter by spatial precision. The L4B neighbourhood centroids were computed as the mean of all MULTIPOLYGON WKT vertex coordinates from the official City of Edmonton boundary file (City_of_Edmonton_-_Neighbourhoods_20260322.csv, 407 neighbourhoods).

3.5. Variable Dictionary

The final modeling dataset (df_final) contains 30 variables classified by source: Original (raw sold CSV), Engineered (transformation pipeline), or External (outside the sold CSV). Each variable is mapped to the research question it supports.

Part A — Identifiers and Core Transaction Variables

Variable	Type	Source	Description	RQ
Account_Number	String	Engineered	Standardized LINC/Account ID after drift reconciliation	RQ1
Base_Address	String	Engineered	Physical parcel key after unit/suite modifier stripping	RQ1
Community_Reconciled	String	Engineered	Canonical neighbourhood name after Oliver→Wihkwêntôwin crosswalk	RQ1/2
Sold_Date	Datetime	Original	Transaction date validated as datetime64[ns]. 18 corrupt rows removed	All
Sale_Year	Integer	Engineered	Calendar year extracted from Sold_Date	All
Sold_Price	Float	Original	Nominal transaction price in CAD. Range: \$1,000–\$18.4M. Median: \$305,000	All
List_Price	Float	Original	Original asking price. Assessed_Value proxy when assessment CSV absent	RQ2
DOM	Integer	Original	Days on market before sale. Median: 32 days	RQ2

Part B — Property Characteristics

Variable	Type	Source	Description	RQ
FlrArea_SF	Float	Original	Above-grade floor area in square feet. Median: 1,221 sqft	RQ2/3
Yr_Built	Integer	Original	Year of original construction	RQ2
Bedrms_AG	Integer	Original	Bedrooms above grade. Mean: 2.59	RQ2
Full_Baths	Integer	Original	Number of full bathrooms. Mean: 1.95	RQ2
Asset_Type	Category	Engineered	Completed Residential (484,028) / Pre-Construction/Land (8,765)	RQ2/3
Price_Per_SQFT_Clean	Float	Engineered	Sold_Price / FlrArea_SF. Null for ghost assets. Median: \$238.34/sqft	RQ2

Part C — Engineered Features (F1-F5)

Variable	Type	Source	Description	RQ
Price_Ratio	Float	Engineered	[F1] Sold_Price / List_Price. Mean: 1.08, Median: 0.97	RQ2
Primary_Driver	Category	Engineered	[F1] Macro if ≤ 1.25 , else Internal. Pearson R = -0.24	RQ2
Rolling_3Yr_Neighbourhood_Growth	Float	Engineered	[F2] 36-month rolling median % change. Range: -53.8% to +233.0%	RQ1/2
Market_Cycle_Phase	Category	Engineered	[F3] Eight named BoC rate regimes (2000–2025)	RQ2/3
Macro_Context	Category	Engineered	[F3] High Rate (94,908) / Low Rate (362,392) / Current Market (57,804)	RQ2/3
Is_Repeat_Transaction	Boolean	Engineered	[F4] True if prior sale at same Base_Address. 370,159 records (75.1%)	RQ1
Days_Since_Last_Sale	Float	Engineered	[F4] Calendar days between repeat sales. Null for first-known	RQ1
Appreciation_Since_Last_Sale_Pct	Float	Engineered	[F4] % price change between repeat sales. Capped $\pm 500\%$. Median: +4.0%	RQ1
Inflation_Adjusted_Price	Float	Engineered	[F5] Sold_Price in constant 2024 CAD. Median: \$383,671	RQ3
Real_Price_Index	Float	Engineered	[F5] Inflation_Adjusted_Price / dataset median $\times 100$. Median = 100	RQ3

Part D — Macroeconomic and Spatial Variables

Variable	Type	Source	Description	RQ
Mortgage_Rate	Float	External	BoC 5-yr rate (V80691335) merged via merge_asof. Range: 4.64%–8.75%. 0 nulls	RQ2/3
Assessed_Value	Float	External	Municipal assessment value. Proxied by List_Price when assessment CSV absent	RQ2
Latitude	Float	External	Geographic latitude. 100% populated via 5-layer pipeline. Geo_Confidence documents tier	RQ3*
Longitude	Float	External	Geographic longitude. 100% populated. Edmonton bounds: -113.71 to -113.33 W	RQ3*
Geo_Source	String	Engineered	Which geocoding layer assigned coordinates (L1_LINC_Exact through L5_CityDefault)	RQ3*
Geo_Confidence	String	Engineered	Spatial accuracy tier: HIGH / MEDIUM / LOW / IMPUTED	RQ3*

4. Data Analysis

4.1. Data Preparation and Collection

The primary dataset was provided as a compressed ZIP archive containing the Edmonton residential sold transactions file (2000–2025). The dataset was extracted using the Python zipfile library and loaded into pandas using the latin1 encoding parameter to prevent character corruption from multi-system data entry. The low_memory=False parameter ensured full column-type inference before assignment.

Potential Sampling Bias: The dataset reflects only completed market transactions recorded at point of sale. Off-market sales, foreclosures, and intra-family transfers are systematically absent. LINC identifier drift introduces a bias toward shorter apparent property histories — directly addressed by the Base_Address reconciliation pipeline.

Data Type and Granularity: The dataset is transaction-level: one row represents one residential property sale. Columns include identifiers, property attributes, pricing, temporal variables, and administrative classification fields. Granularity is daily (Sold_Date) at the individual property level.

4.2. Data Transformation

The Oliver – Wihkwêntôwin Problem

The historic Edmonton neighbourhood known as Oliver was officially renamed Wihkwêntôwin. Without correction, time-series analysis treats the neighbourhood as two separate geographic regions. A categorical crosswalk dictionary was implemented to map legacy neighbourhood names to modern equivalents, updating 9,357 records and ensuring all historical records are grouped under a consistent identifier.

LINC Identifier Drift

LINC drift was addressed through Base_Address engineering, which strips unit modifiers such as 'Unit A' or 'Suite 102' from address strings. This links multiple administrative identifiers to the same physical land parcel. The process resolved 234,486 drift instances (65.66% efficiency), identifying 122,634 unique physical parcels from 357,120 raw administrative identifiers.

4.3. Data Engineering

Ghost Asset Isolation

Exploratory analysis identified 9,246 records where properties sold for more than \$100,000 but reported floor areas of zero or one square foot — pre-construction sales. Including them creates extreme price-per-square-foot outliers. These records were retained but reclassified as Pre-Construction/Land.

Macroeconomic Synchronization

Historical mortgage rate data from the Bank of Canada (Series V80691335) was loaded from the official BoC CSV, providing 1,362 weekly observations spanning 2000–2026. The data was merged using pandas merge_asof, aligning each property sale with the most recent interest rate available before the

transaction date — ensuring temporal consistency with zero forward leakage.

4.4. Exploratory Data Analysis

Correlation analysis confirmed that the strongest bivariate relationships with Sold_Price are floor area (positive, non-linear), Sale_Year (positive — long-run trend), and Mortgage_Rate (negative — Pearson $R = -0.238$ in the Macro segment). The Rolling 3-Year Neighbourhood Growth feature showed a range of -53.8% to +233.0%, confirming substantial local-level heterogeneity beyond the macro signal.

The nominal price distribution is strongly right-skewed with a luxury tail. Log transformation was applied before OLS regression. In constant 2024 dollars, nominal prices rose from \$145,500 (2000) to \$424,000 (2025) — a +189% nominal gain. After CPI deflation (Alberta CPI, Stats Can 18-10-0004-01), real prices rose +73%, confirming approximately 60% of apparent appreciation was general inflation.

Variable	Min	Q1	Q3	Max	Action
Sold_Price	\$1,000	\$305,000	\$18.4M	0.06%	Retained — valid luxury transactions
Price_Ratio	0.05	0.97	>20.0	< 1%	Capped at 20.0 for analysis
Mortgage_Rate	4.64%	5.88%	8.75%	0%	Zero nulls after fillna(8.25%)
FlrArea_SF	0	1,221 sqft	12,000+	1.8%	Ghost assets reclassified
DOM	0	32 days	999+	0.4%	Retained — extreme listings valid

5. Data Modeling and Inferences

5.1. Machine Learning Problem Type

This project addresses a **supervised regression problem**. The target variable — `Sold_Price` — is a continuous numeric value. Regression was selected over classification because predicting an exact dollar amount is more informative for decision-support purposes and allows direct error measurement via MAE and RMSE.

5.2. Models Selected and Justification

Three models of increasing complexity were trained to allow comparative evaluation and to demonstrate progressive performance improvement.

OLS Linear Regression: Interpretable baseline. Coefficient significance is directly testable via t-statistics, providing inferential insight alongside predictive accuracy. Assumes linear relationships between features and price.

Random Forest (200 trees, max_depth=15, min_samples_leaf=10): Captures non-linear interactions between floor area, neighbourhood, and rate cycles without explicit feature engineering. Reduces variance by averaging 200 independently trained trees.

Gradient Boosting (200 estimators, lr=0.05, subsample=0.8): Best model. Reduces bias through sequential residual correction — each tree corrects the errors of the previous one (stagewise fitting). Shrinkage (learning_rate=0.05) prevents any single tree from overfitting on noise. Subsampling (0.8) acts as regularization. Gradient Boosting outperforms Random Forest on structured tabular data with systematic economic patterns because bias reduction is more valuable than variance reduction when the signal is structured rather than random.

5.3. Feature Engineering Rationale

Sixteen features were selected spanning four categories. All categorical variables were label-encoded before model fitting.

Category	Features	Rationale
Macroeconomic	Mortgage_Rate, Context_Encoded, Phase_Encoded, Driver_Encoded	Captures interest rate ceiling effect (Pearson $R=-0.238$). Phase encoding allows model to distinguish 2022 Rate Shock from 2010 Low Rate era
Temporal	Sale_Year, Sale_Month, Is_Spring_Market	Sale_Year captures 25-year trend. Sale_Month and Is_Spring_Market capture Edmonton's pronounced spring market peak (March–June)
Property	Log_FlrArea, Bedrms_AG, Full_Baths, DOM, Property_Age	Log transform handles non-linear price-size relationship. DOM is a market liquidity signal — high DOM in high-rate periods confirms macro suppression

Feature Group	Features	Description
Engineered (F1-F5)	Rolling_3Yr_Growth, Real_Price_Index, Days_Since_Last_Sale, Appreciation_%	Directly embeds the five pre-modeling features, connecting pipeline outputs to model inputs

5.4. Cross-Validation Approach

Standard k-fold cross-validation was deliberately not applied. Shuffling observations randomly allows future months to appear in training folds, introducing temporal leakage — the exact problem the pre-modeling pipeline was designed to prevent. A temporal holdout split was used instead:

- Training set: 418,302 records from January 2000 to December 2023
- Test set: 54,818 records from January 2024 to December 2025
- This simulates the realistic deployment scenario of predicting future prices from past data only

This is the methodologically correct evaluation strategy for time-series regression. The two-year test period (2024–2025) provides a robust evaluation window that includes seasonal variation and market normalization dynamics.

5.5. Bias-Variance Tradeoff and Overfitting Assessment

VIF analysis revealed extreme multicollinearity among trend features: Sale_Year (VIF=560.51) and Real_Price_Index are collinear representations of the long-run trend. While coefficient estimates remain unbiased, individual importance scores overstate unique contributions.

The Gradient Boosting training R^2 of 0.9998 versus test R^2 of 0.9922 indicates mild overfitting — the model memorizes some training patterns absent in the test period. The gap is acceptable for a draft model. Residual heteroskedasticity was confirmed: MAE grows from \$7,990 in the <\$300K band to \$59,865 in the >\$1M band. Percentage error (MAPE = 3.92%) remains stable, indicating the model is well-calibrated in proportional terms. OLS diagnostics confirmed non-normal residuals (Shapiro-Wilk $W=0.931$, $p<0.001$) and heteroskedasticity (Breusch-Pagan $p<0.001$) — expected for 278,449-record regression on right-skewed price data.

	MAE (\$)	RMSE (\$)	R^2	MAPE (%)	Within 5%	Within 10%
MAE (\$)	\$35,206	\$20,996		\$16,536		
RMSE (\$)	\$50,447	\$29,832		\$20,343		
R^2	0.9518	0.9831	0.9922			
MAPE (%)	15.28%	4.73%	3.92%			
Within 5%	41.3%	78.4%	83.0%			
Within 10%	67.8%	98.1%	99.9%			

5.6. Inferential Analysis: RQ1, RQ2, RQ3

RQ1 — Statistical Validation: A Repeat-Sale Price Index was constructed using 88,342 repeat transactions (after 180-day holding period filter). Reconciled RSI std = 17.34% vs broken RSI std =

13.49%. Levene's test ($F=1.33$, $p=0.25$) did not reach significance at 5% — attributable to the small number of quarterly RSI data points rather than absence of a real effect. Bootstrap 95% CIs: reconciled [11.87%, 21.47%], broken [8.96%, 17.16%]. The higher volatility in the reconciled RSI reflects genuine cross-cycle appreciation swings.

RQ2 — Hypothesis Test: H_0 : prices do not differ across market cycle phases. H_1 : at least one phase differs. $\alpha = 0.05$. One-way ANOVA: $F = 16,194.68$ ($p < 0.001$). Reject H_0 . Cohen's $f = 0.484$ (large effect). Tukey HSD post-hoc: all 28 pairwise comparisons between the 8 phases are statistically significant. Pearson $R = -0.238$ ($p < 0.001$) in the Macro segment. OLS Mortgage_Rate coefficient = -0.116 ($p < 0.001$). Endogeneity limitation acknowledged: mortgage rates treated as exogenous; BoC rate decisions may partially respond to housing market conditions.

RQ3 — Stationarity and Model Selection: ADF test confirmed non-stationarity at level ($p=0.164$ for Model B), stationary at first difference ($p=0.003$), confirming $d=1$. AIC grid search over SARIMA($p,1,q$)($1,1,1,12$) for $p,q \in \{0,1,2\}$ identified (2,1,2) as optimal (AIC=5,452.71). Ljung-Box test (lag 12) $p=0.88$ for Model A confirms no residual autocorrelation. Both models outperform the random walk benchmark, confirming SARIMA adds genuine forecasting value.

5.7. Model Improvement Opportunities

- Remove Sale_Year from the feature set and retain Real_Price_Index to eliminate the $VIF=560$ collinearity while preserving the trend signal
- Implement systematic hyperparameter tuning (GridSearchCV or Bayesian optimization) — current parameters were manually set for the draft submission
- Apply single-family Prop_Class filter before using repeat-sale features to prevent condo over-matching distortion
- Add spatial proximity features (distance to LRT stations, green space, schools) using HIGH/MEDIUM confidence geocoded records only
- Retrain annually as new transaction data becomes available — the model generalizes within rate regimes it has seen but may not extrapolate to novel structural breaks

6. Results and Findings

6.1. RQ1 — Reconciliation Findings

The reconciliation pipeline successfully restored historical continuity across the dataset. Of 515,104 valid transactions, 370,159 (75.1%) were confirmed as repeat sales of the same physical property. The Repeat-Sale Price Index revealed that reconciled property chains produce higher apparent appreciation volatility — not because reconciliation introduces noise, but because longer chains correctly capture price movements across multiple economic cycles. The top neighbourhood by median inter-sale appreciation was Willow Park at +44.9%, followed by Heritage Point at +41.6%. Median appreciation across all repeat transactions was +8.89% per sale (mean +22.54%, std dev 50.69%).

6.2. RQ2 — Price Driver Findings

Macroeconomic forces dominate Edmonton residential pricing. 490,487 transactions (99.5%) are Macro-Driven with $\text{Price_Ratio} \leq 1.25$. Pearson $R = -0.238$ in the Macro segment confirms interest rates act as a price ceiling. OLS regression confirmed all features are statistically significant at $p < 0.001$ with $R^2 = 0.524$. The most pronounced price difference is between the Post-Dot-Com High Rate era (\$130,000 median) and the Normalizing Market (\$410,000 median) — a \$280,000 differential explained primarily by the rate decline from 8.25% to 6.50%.

Period	Transactions	Median Price	Interest Rate
Post-Dot-Com (High Rate) 2000–02	45,330	\$130,000	7.60%
Pre-Crisis Expansion 2003–07	95,195	\$204,900	6.46%
Financial Crisis Rate Cut 2008–09	37,019	\$312,500	6.34%
Recovery & Low Rate Era 2010–14	94,228	\$327,500	5.26%
Oil Shock & Stability 2015–19	91,603	\$349,000	4.91%
Pandemic Era (Ultra-Low) 2020–21	44,347	\$351,000	4.84%
Rate Shock Era (High Rate) 2022–23	49,578	\$367,000	6.04%
Normalizing Market 2024–25	57,804	\$410,000	6.50%

6.3. RQ3 — Forecast Accuracy Findings

The AIC-optimal SARIMA(2,1,2)(1,1,1,12) specification outperformed the initial (1,1,1) order by 47.6 AIC points. Model B (Reconciled) underperforms Seasonal Naïve in the stable 2024–2025 test period — an honest finding: when prices are near-flat, year-over-year patterns are highly competitive. Both models outperform the random walk benchmark. The 2026 forecast (Model B, constant 2024\$) projects a January value of \$415,555 rising to a mid-year peak near \$448,260, with 95% confidence intervals widening to approximately $\pm \$96,000$ by May 2026.

6.4. Dataset Validation Results

Validation Check	Criterion	Result	Status
Critical variable completeness	< 5% null	8/9 at 0.00% Community_Reconciled: 3.44%	PASS
Datetime integrity	datetime64, no NaT	0 nulls; range 2000–2025	PASS
Sold_Price distribution	\$10K–\$5M	Median \$305,000; 0.06% outliers	PASS
Price_Ratio distribution	0.05–20.0	Median 0.97; < 1% outliers	PASS
Mortgage_Rate range	2.0%–12.0%	Range 4.64%–8.75%; 0 nulls after fill	PASS
Appreciation capped	±500%	Max 500.0%; 1,618 values clipped	PASS
Final dataset shape	—	492,793 rows × 30 columns	PASS
CSV exported	—	Edmonton_RealEstate_ModelingReady_df_final_G5.csv	PASS

7. Business Implications and Recommendations

7.1. For Policymakers and City Planners

The finding that macroeconomic forces dominate 99.5% of Edmonton transactions has direct policy implications. Interest rate decisions made at the national level function as an effective price ceiling for the local housing market. The 2022–2023 rate shock era produced a \$367,000 median transaction price despite a strong economy — suppressing prices that would otherwise have continued rising.

- Municipal policy interventions should be timed relative to rate cycles. Infill development incentives are most effective during high-rate periods when buyer purchasing power is constrained
- Neighbourhood-level growth rates (Feature F2) identify areas where local structural factors are outpacing the macro ceiling — these are priority areas for infrastructure investment and zoning review
- The 2026 forecast range of \$415K–\$448K (constant 2024\$) suggests continued normalization — affordable housing programs targeting the sub-\$350K segment remain relevant

7.2. For Real Estate Investors and Developers

The Gradient Boosting model ($R^2 = 0.9922$, MAE = \$16,536) provides a reliable price prediction tool for individual transactions within the Completed Residential segment. 83.0% of predictions are within 5% of actual price, and 99.9% are within 10%. This accuracy level is sufficient for acquisition screening, portfolio valuation, and development feasibility assessments.

- The model underperforms on luxury properties (>\$1M, MAE = \$59,865 vs \$7,990 for <\$300K properties) — apply with caution in the high-end segment
- The Rolling_3Yr_Neighbourhood_Growth feature (Feature F2) is the cleanest available signal of locally-driven appreciation — neighbourhoods in the top quartile (+28%+) have outperformed the macro ceiling
- Pre-construction transactions are excluded from the model — the 9,246 ghost assets represent a separate analysis track for development-stage investments

7.3. For Data Practitioners and Technical Stakeholders

This project demonstrates that investing in data reconciliation before model development produces compounding returns. The 234,486 resolved LINC drift instances enabled 370,159 repeat transaction chains that would otherwise be invisible to any model. This is not merely a data cleaning exercise — it is a structural correction that transforms the analytical capability of the entire dataset.

- The 5-layer geocoding pipeline achieves 100% spatial coverage in seconds — a practical replacement for API-based geocoding at any scale above 10,000 records
- The Geo_Source and Geo_Confidence metadata columns enable transparent spatial analysis: filter to HIGH/MEDIUM confidence records for point-level analysis, include LOW for neighbourhood-level
- Temporal holdout evaluation is mandatory for time-series regression — k-fold cross-validation introduces data leakage and produces artificially high accuracy metrics
- The VIF=560 finding on Sale_Year is a caution for any real estate ML practitioner: high R^2 in time-series regression frequently reflects trend extrapolation rather than causal understanding

8. Analysis Limitations and Constraints

8.1. Collinearity in Feature Set

VIF analysis confirmed extreme multicollinearity: Sale_Year (VIF=560.51) and Real_Price_Index are collinear representations of the long-run price trend. While coefficient estimates remain unbiased, individual feature importance scores overstate each feature's unique contribution. The model's high R^2 partially reflects trend extrapolation rather than structural price understanding.

Mitigation: Remove Sale_Year from the final model feature set and evaluate performance change. Retain Real_Price_Index as the inflation-adjusted trend signal.

8.2. Algorithmic Over-Matching in Base_Address Engineering

The strategy of stripping unit and suite modifiers introduces over-matching risk in high-density developments where distinct legal parcels share a single civic address. Unit A and Unit B at the same building may be incorrectly linked as repeat sales of the same property.

Mitigation: Restrict repeat-sale analysis to single-family detached Prop_Class records where the risk is minimal.

8.3. Temporal Distribution Shift

The 2024–2025 test period falls entirely within the Normalizing Market phase. The model has not been tested on a rate-shock or financial crisis scenario using the current feature set. Its ability to generalize to structural breaks not seen during training is unknown.

Mitigation: Retrain annually. Include new rate cycle phases as they develop. Monitor prediction accuracy by quarter.

8.4. SARIMA Benchmark Performance

SARIMA Model B (Reconciled) underperforms the Seasonal Naïve benchmark in the 2024–2025 stable market (MAE \$47,260 vs \$23,652). This is an honest and acknowledged result. SARIMA adds forecasting value in volatile periods but is difficult to justify over year-over-year naïve forecasting in near-flat market conditions.

Mitigation: Consider hybrid models (SARIMA + exogenous BoC rate regressors via SARIMAX) for the final submission.

8.5. Geocoding Confidence

Approximately 83,148 records (16.1%) carry IMPUTED geocoding (Edmonton city centre default). These records are not suitable for point-level spatial analysis and were excluded from the ML model's spatial features.

Mitigation: Filter to HIGH and MEDIUM confidence records for any proximity or spatial modelling.

8.6. Temporal Leakage in Rolling Growth Feature

Rolling_3Yr_Neighbourhood_Growth was computed on the full 25-year dataset before any train/test split. Test-period records may draw from future months in the rolling calculation.

Mitigation: Recompute the rolling feature on training data only before fitting the final model.

8.7. Source Data Integrity

The dataset contains inherited noise from original manual entries, including typographic errors in street names and inaccuracies in transaction dates. While the pipeline filters extreme outliers, minor clerical errors remain a persistent source of variance.

Mitigation: Extreme outlier filtering applied. Residual clerical errors acknowledged as irreducible noise.

9. Conclusion

This project demonstrates that data integrity is a foundational requirement for accurate real estate forecasting. Rather than focusing solely on modelling techniques, the study emphasizes the importance of repairing fragmented datasets before applying predictive algorithms. The eight-stage transformation pipeline converted 515,122 raw transactions into a 492,793-record, 30-column modeling-ready dataset by resolving 234,486 LINC drift instances, isolating 9,246 ghost assets, reconciling neighbourhood naming, integrating 1,362 weekly Bank of Canada mortgage rate observations with zero forward leakage, and implementing a 5-layer offline geocoding pipeline achieving 100% spatial coverage.

The Gradient Boosting model achieved $R^2 = 0.9922$ and MAE = \$16,536 on the 2024–2025 holdout set, with 83.0% of predictions within 5% of actual price. The high R^2 is partly attributable to collinear trend features and should be interpreted with caution — predictive accuracy does not equal explanatory insight.

All three research questions were answered with supporting statistical evidence. RQ1 confirmed that reconciliation restores repeat transaction chains and changes RSI curve behaviour. RQ2 confirmed that macroeconomic forces dominate (Pearson $R = -0.238$, $F = 16,194$ across phases, large effect size; all 28 post-hoc pairs significant). RQ3 confirmed that the AIC-optimal SARIMA(2,1,2)(1,1,1,12) adds forecasting value over the random walk benchmark, with a 2026 real-price forecast of \$415,555–\$448,260.

All deliverables described in this report — the pre-modeling notebook, enhanced modeling notebook, output CSV, executive summary, ML process summary, Streamlit dashboard, and Power BI guide — are consistent with and reproducible from the artifacts submitted alongside this document.

10. References

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